Examiner only

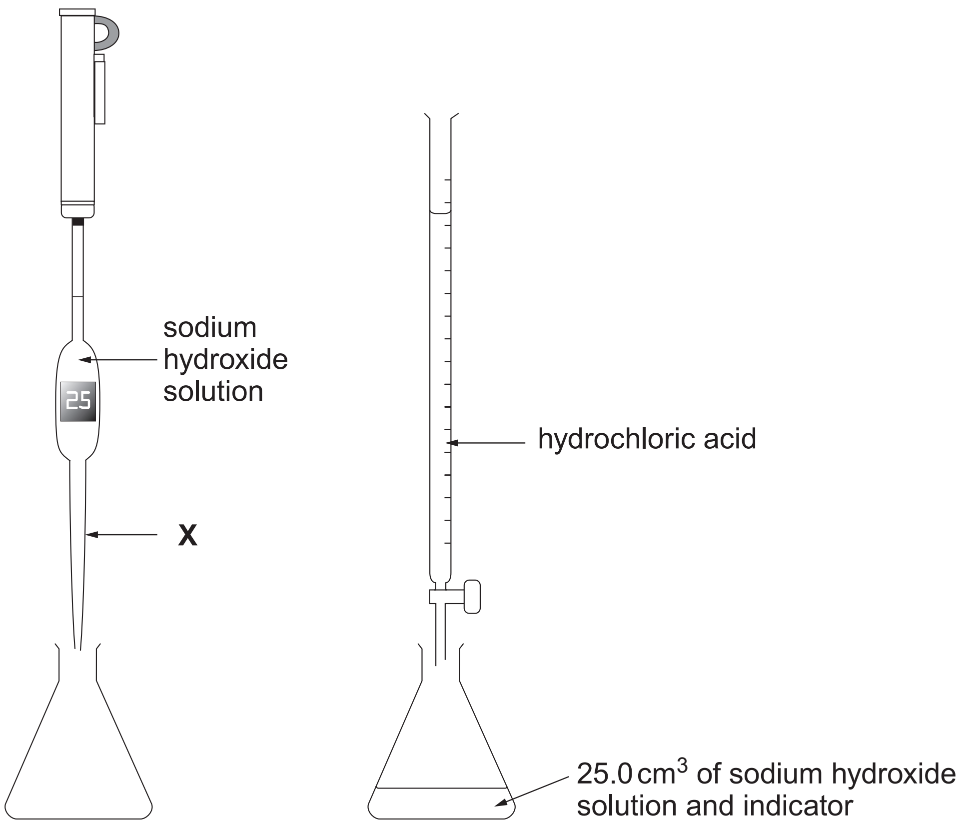
7. *(a)* Hydrochloric acid, HCl, reacts with sodium hydroxide solution to form sodium chloride and water only.

Write a balanced **symbol** equation for this reaction. [2]

....

(b) A group of students was asked to find the volume of hydrochloric acid solution needed to neutralise 25.0 cm^3 of sodium hydroxide solution. They decided to titrate sodium hydroxide with hydrochloric acid.

Apparatus



Results

Titre	1	2	3
Volume of hydrochloric acid needed (cm^3)	18.2	17.8	18.0

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(i) Name the piece of apparatus **X**. [1]

....

(ii) Explain the purpose of the indicator. [1]

....

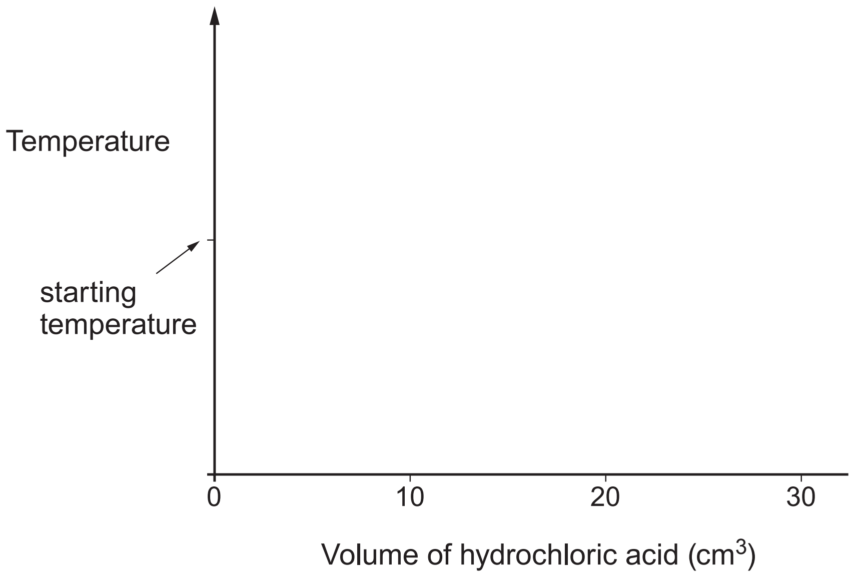
....

(iii) Calculate the mean volume of hydrochloric acid needed to neutralise 25.0 cm^3 of the sodium hydroxide solution. [1]

Mean volume = ... cm^3

(iv) The change in temperature during the reaction can be monitored using a temperature sensor.

Sketch a graph on the axes below to show how the temperature changes as more and more acid up to a total of 30 cm^3 is added. [2]



(v) The experiment was repeated using hydrochloric acid of **half** the original concentration.

State the volume of hydrochloric acid that would be needed to change the indicator colour. [1]

Volume = ... cm^3

8

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